Atomic Static Dipole Polarizability Generative Model

Development Booklet: Parts A–D

Descriptor Design · Fitting Methodology · Validation · Plateau Investigation

Formation Engine — Atomistic Simulation Library

January 2025

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Purpose

This booklet is the single authoritative record for the atomic static dipole polarizability generative model (the *alpha model*) built into the Formation Engine atomistic simulation library. The model predicts the scalar isotropic static dipole polarizability $\alpha(Z)$ for neutral atoms $Z = 1$–120 using only the atomic number as input; no external data files are read at runtime. The predicted values feed directly into the Thole-damped SCF induced-dipole solver that computes polarization energies and forces during molecular dynamics.

The document is organised in two parts reflecting the chronological development record:

Part I **Model Design and Validation (Parts A and B).** Derivation of the model equation, physical motivation for each term, descriptor definitions, fitting methodology, the v2.6.0–v2.6.2 parameter progression, validation results, integration with the SCF solver, nuclear stability integration, and the software architecture. This constitutes the primary scientific contribution.

Part II **Plateau Investigation (Part C).** Two incremental parameter additions—v2.6.3 and v2.7.0—were attempted to push the global RMS below 15%. Both failed to break the 17% floor. This part documents the experiments, a four-cause root-cause analysis of why the floor exists, and a decision log that reverts the active codebase to v2.6.2. The anti-pattern log records every approach that has been tried and failed so that future development targets root causes rather than symptoms.

The active production code is **v2.6.2** (15 parameters, global RMS 17.5%). Its failure modes are fully characterised and should be read before any further model development is attempted.

Part I

Model Design and Validation

Parts A and B — v2.6.0 through v2.6.2

Chapter 1

Model Equation

Abstract

A 16-parameter generative model for the scalar isotropic static dipole polarizability $\alpha(Z)$ of neutral atoms covering $Z = 1$–118 with an experimental extension to $Z = 120$. The model requires no external data files at runtime: all inputs are derived from the atomic number Z alone via closed-form descriptor functions (covalent radius, period, block, electronegativity proxy, softness). Parameters are fitted offline against 118 reference values from Miller (1990) and Schwerdtfeger & Nagle (2019) using iteratively-reweighted Huber regression with group-monotonicity regularisation. The overall RMS error is 19.1% across all 118 elements and <16% for the commonly-simulated $Z = 1$–54 subset. The model feeds directly into the Thole-damped SCF induced-dipole solver that computes polarization energies and forces in molecular dynamics.

1.1 Equation

The polarizability of element Z is predicted by

$$\boxed{\alpha(Z) \;=\; \underbrace{c_b(Z)\, k_p(Z)\, r_{\mathrm{cov}}(Z)^3\, \frac{1 + a_{\mathrm{soft}}\, s(Z)}{\chi(Z)^{b_\chi}}}_{\alpha_{\mathrm{base}}} \cdot \underbrace{g_{\mathrm{shell}}(Z)}_{\text{closure gate}} \cdot \underbrace{g_{\mathrm{rel}}(Z)}_{\text{relativ. stiffening}}} \tag{1.1}$$

where the terms are:

Symbol	Meaning	Source
$r_{\mathrm{cov}}(Z)$	Covalent radius (Å)	Pyykkö & Atsumi (2009)
$s(Z)$	Softness descriptor (dimensionless)	Eq. (4.4)
$\chi(Z)$	Electronegativity proxy	Eq. (4.3)
$k_p(Z)$	Period scaling factor ($p = 1 \ldots 7$)	Fitted
$c_b(Z)$	Block correction ($b \in \{s, p, d, f\}$)	Fitted
a_{soft}	Softness coupling coefficient	Fitted
b_χ	Electronegativity suppression exponent	Fitted
$g_{\mathrm{shell}}(Z)$	Shell-closure suppression (Eq. 2.2)	Fitted
$g_{\mathrm{rel}}(Z)$	Relativistic stiffening (Eq. 3.2)	Fitted

The model has **15 free parameters**: 7 period coefficients $k_p^{(1)} \dots k_p^{(7)}$, 4 block corrections $c_b^{(s)}, c_b^{(p)}, c_b^{(d)}, c_b^{(f)}$, 2 global exponents a_{soft}, b_χ, and 2 gate amplitudes $c_{\text{shell}}, a_{\text{rel}}$.

1.2 Physical Motivation

The leading-order physics is the *volume scaling law*: polarizability grows as the cube of the electron-cloud radius. For neutral atoms the covalent radius r_{cov} is the best single-descriptor proxy for the mean valence-shell extent. The remaining terms correct for effects that r_{cov}^3 alone cannot capture:

1. **Period scaling (k_p):** Outer-shell electrons in higher periods are more diffuse per unit radius due to additional radial nodes. The fitted k_p values decrease monotonically from period 1 to period 7, reflecting the diminishing "volume efficiency" as shielding increases.
2. **Electronegativity suppression (χ^{b_χ}):** Electronegative atoms hold their electron cloud tighter, reducing deformability below what r_{cov}^3 predicts. The denominator χ^{b_χ} implements this as a power-law suppression.
3. **Softness coupling ($1 + a_{\text{soft}}\, s$):** The softness descriptor s measures how loosely-held the valence electrons are (high for alkali metals, low for noble gases). The fitted $a_{\text{soft}} < 0$ *reduces* α for soft atoms—a counter-intuitive result that compensates for the period scaling already being fit to alkali metals.
4. **Block correction (c_b):** Transition metals (d-block) and lanthanides/actinides (f-block) have polarisable sub-valence electrons ($3d$, $4f$) whose spatial extent is not captured by r_{cov}. The block multiplier absorbs this orbital-type bias.

Chapter 2

Shell-Closure Gate

2.1 Why Noble Gases Blow Up (and How to Fix Without Hacks)

Noble gases are closed-shell atoms: their electron clouds are stiff and compact. Polarizability is comparatively low and follows a *different curvature* than smooth volume scaling—shell closure breaks the monotonic "bigger radius ⇒ more polarisable" intuition that the base model assumes.

Before v2.6.2, the noble-gas RMS was 36.7%, with Ne at +85%. The existing χ-based denominator in α_{base} partially corrected for this (by assigning noble gases a high effective electronegativity), but was insufficient because r^3_{cov} still dominates.

2.2 The Shell-Closure Gate

Noble gases should not be "smoothed" into the same curve as their neighbours. They must be *gated*: a multiplicative penalty that activates at or near shell closure.

We define a *shell-closure proximity* descriptor:

$$\phi(Z) = \exp\left(-\frac{(18 - g(Z))^2}{2\sigma^2_{\text{shell}}}\right) \tag{2.1}$$

where $g(Z)$ is the IUPAC group number ($g = 18$ for noble gases) and $\sigma_{\text{shell}} = 1.8$. This Gaussian is centred on group 18:

- Noble gases ($g = 18$): $\phi = 1.0$ (full closure)
- Halogens ($g = 17$): $\phi \approx 0.86$ (near-closure)
- Chalcogens ($g = 16$): $\phi \approx 0.55$ (moderate)
- $g \leq 14$: $\phi < 0.15$ (negligible)
- f-block ($g = 0$): $\phi = 0$ by convention

The gate is then:

$$g_{\text{shell}}(Z) = 1 - c_{\text{shell}} \cdot \phi(Z) \tag{2.2}$$

where c_{shell} is a single fitted amplitude. The fitted value $c_{\text{shell}} = 0.328$ means noble-gas polarizabilities are suppressed by $\sim 33\%$ relative to the base model, while halogens get a $\sim 28\%$ suppression—physically correct, as the halogen anion X^- (iso-electronic with the adjacent noble gas) has far higher polarizability than neutral X.

Design principle: gate, don't smooth. Smoothing residuals across a window that includes noble gases will try to "average out" closure physics, creating a correction curve that looks pretty and is physically wrong. Noble gases are a *regime boundary*: the model must acknowledge the discontinuity in electronic structure at shell closure, not paper over it.

Chapter 3

Relativistic Stiffening

3.1 Why Mercury Spikes (and What Fixes It)

Mercury ($Z = 80$) is the poster child for relativistic orbital contraction: its $6s$ orbital contracts and stabilises, reducing participation in polarizability and making Hg unusually "inert" compared to naive trend expectations. This is the same physics behind liquid mercury and its anomalous bonding.

Before v2.6.2, the model treated Hg like a normal heavy metal with a fat, squishy outer cloud: predicted 7.79 Å^3 vs. reference 5.70 Å^3 (+37%).

3.2 The Relativistic Stiffening Factor

Relativistic contraction is *not* monotonic with Z—it peaks at specific electronic configurations. We identify two hotspots:

1. $6s$ **inert-pair region** ($Z \approx 80$): Au(79), Hg(80), Tl(81) all have anomalously compact $6s$ orbitals due to strong nuclear shielding.
2. $7p_{1/2}$ **spin–orbit region** ($Z \approx 115$): The superheavy p-block (Nh–Og) experiences extreme j-dependent orbital splitting. The $7p_{1/2}$ subshell contracts strongly, reducing the effective polarisable volume.

We model these as a sum of two Gaussians:

$$f_{\text{rel}}(Z) = \exp\left(-\frac{(Z-80)^2}{200}\right) + \frac{1}{2}\exp\left(-\frac{(Z-115)^2}{50}\right) \tag{3.1}$$

and the gate is:

$$g_{\text{rel}}(Z) = 1 - a_{\text{rel}} \cdot f_{\text{rel}}(Z) \tag{3.2}$$

The first Gaussian ($\sigma = 10$) yields $f_{\text{rel}} \approx 1.0$ at Hg, ≈ 0.7 at Au/Tl, and decays to < 0.1 by Gd(64) and Po(84). The second Gaussian ($\sigma \approx 5$) peaks at 0.5 for Mc(115) and covers Nh(113)–Og(118).

The fitted amplitude $a_{\text{rel}} = 0.054$ is physically small, consistent with the expectation that relativistic effects are a *perturbative correction* to the volume-scaling base model, not a dominant term. The correction reduces Hg's error from +37% to +34% and improves the overall period-6 RMS from 15.4% to 14.1%.

Why not a sigmoid? An earlier version used a sigmoid $\sigma(Z) = 1/(1 + e^{-(Z-76)/8})$. This failed ($a_{\mathrm{rel}} = 0$ in all fits) because it applies uniform dampening to *all* heavy atoms: Os, Ir, and Pt—which fit well without correction—were degraded more than Hg was improved. The double-Gaussian targets only the physical hotspots, allowing the fitter to find a non-zero optimum.

Hg as a calibration anchor. If Hg's error improves, Au, Rn, and the $7p$ "inert pair" elements (Nh–Og) typically improve as well, because the same $6s/7p_{1/2}$ contraction physics underlies all of them. Hg should therefore be monitored as a leading indicator in future model revisions.

Chapter 4

Descriptor Definitions

All descriptors are pure functions of Z. No external database is loaded at runtime.

4.1 Covalent Radius $r_{\mathrm{cov}}(Z)$

Tabulated for $Z = 1$–118 from Pyykkö & Atsumi, *Chem. Eur. J.* **15**, 186 (2009). Stored as a compile-time constant array. Units: Å.

4.2 Period and Block

Standard IUPAC periodic table assignment:

$$p(Z) \in \{1, 2, 3, 4, 5, 6, 7\}, \qquad b(Z) \in \{s,\, p,\, d,\, f\}$$

Lanthanides ($Z = 57$–71) and actinides ($Z = 89$–103) are assigned $b = f$. Group 0 (no IUPAC group number) is used for the f-block insets; all other elements have standard group assignments $g = 1$–18.

4.3 Active Valence Electron Count $n_{\mathrm{val}}(Z)$

$$n_{\mathrm{val}}(Z) = \begin{cases} g & g \in \{1, 2\} \quad (s\text{-block}) \\ g - 10 & g \in \{13, \ldots, 18\} \quad (p\text{-block}) \\ g & g \in \{3, \ldots, 12\} \quad (d\text{-block}) \\ 3 & Z \in [57, 71] \cup [89, 103] \quad (f\text{-block}) \end{cases} \tag{4.1}$$

f-block design decision. Lanthanide and actinide $4f/5f$ electrons are spatially contracted and behave as core-like. Setting $n_{\mathrm{val}} = 3$ for all 30 elements prevents an artificial monotonic gradient in $s(Z)$ across each series. The actual variation in polarizability within the lanthanide contraction is absorbed by $c_b^{(f)}$ and $r_{\mathrm{cov}}(Z)^3$. This invariant is enforced by test T1.

4.4 Maximum Period Valence $n_{\max}(p)$

$$n_{\max}(p) = \begin{cases} 2 & p = 1 \\ 8 & p \in \{2, 3\} \\ 18 & p \in \{4, 5\} \\ 32 & p \in \{6, 7\} \end{cases} \tag{4.2}$$

4.5 Electronegativity Proxy $\chi(Z)$

An Allred–Rochow-style estimate:

$$\chi(Z) = \frac{n^*_{\text{val}}(Z)}{r_{\text{cov}}(Z)^2} \times 0.359 \tag{4.3}$$

where the effective valence count n^*_{val} equals $n_{\text{val}}(Z)$ for all elements except noble gases ($g = 18$), which use $n^*_{\text{val}} = n_{\max}(p)$ (full shell). The normalisation constant 0.359 is chosen so that carbon ($n_{\text{val}} = 4$, $r_{\text{cov}} = 0.75$ Å) gives $\chi_{\text{C}} \approx 2.55$ on the Pauling scale.

Noble gas electronegativity. Setting $n^*_{\text{val}} = n_{\max}$ for group 18 produces artificially high χ, which drives α down through the denominator of Eq. (1.1). This is physically correct: noble gases have compact, tightly-bound closed shells and small polarizabilities despite moderate covalent radii.

4.6 Softness $s(Z)$

$$s(Z) = \frac{n_{\text{val}}(Z) \;/\; n_{\max}\big(p(Z)\big)}{\chi(Z)} \tag{4.4}$$

Softness is high for diffuse, easily-deformable electron clouds (alkali metals: $s_{\text{K}} \approx 5.5$) and low for tight shells (fluorine: $s_{\text{F}} \approx 0.3$; neon: $s_{\text{Ne}} \approx 0.03$).

Chapter 5

Fitting Methodology

5.1 Reference Data

118 reference polarizabilities from two sources:

- **Miller (1990):** gas-phase experimental values for $Z = 1$–54 (JACS 112, 8533).
- **Schwerdtfeger & Nagle (2019):** relativistic DHF/CCSD(T) values for $Z = 21$–118 (Mol. Phys. 117, 1200).

Each entry carries a fitting weight $w_i \in [0.1, 2.0]$ encoding reliability: $w = 2.0$ for high-priority simulation elements (H, C, N, O, S), $w = 0.3$ for radioactive elements with theory-only values.

5.2 Loss Function

The objective is a weighted Huber loss in log-space plus a smoothness penalty:

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^{N} w_i \, \mathcal{H}_\delta\Big(\ln \hat{\alpha}_i - \ln \alpha_i^{\mathrm{ref}}\Big) \; + \; \eta \sum_{(a,b)\in\mathcal{P}} \mathcal{H}_\delta\Big(\Delta_{ab}^{\mathrm{pred}} - \Delta_{ab}^{\mathrm{ref}}\Big) \tag{5.1}$$

where $\mathcal{H}_\delta(x) = \frac{1}{2}x^2$ for $|x| \le \delta$ and $\delta(|x| - \frac{1}{2}\delta)$ otherwise, with $\delta = 0.3$. The set $\mathcal{P}$ contains pairs of elements that are adjacent in the same periodic-table group, sorted by period; $\Delta_{ab} = \ln \alpha_b - \ln \alpha_a$ enforces monotonicity of trends down each group.

Why log-space. Polarizabilities span three orders of magnitude ($\alpha_{\mathrm{He}} = 0.205$ Å^3 to $\alpha_{\mathrm{Cs}} = 59.4$ Å^3). Log-space Huber loss treats a 50% error on helium as seriously as a 50% error on caesium.

5.3 Optimisation: Iterative Reweighted Coordinate Descent

The 15 parameters are fitted by a two-loop scheme:

1. **Outer loop** (up to 30 rounds): after each inner convergence, update element weights w_i based on residuals:

$$w_i \leftarrow \min\big(w_{\max},\; w_i\big(1 + \alpha_{\mathrm{rw}} \cdot \max(0,\; |\epsilon_i|/\tau - 1)\big)\big)$$

where $\epsilon_i = \ln \hat{\alpha}_i - \ln \alpha_i^{\mathrm{ref}}$, $\tau = 0.20$ is the tolerance target, $\alpha_{\mathrm{rw}} = 0.3$, and $w_{\max} = 10$.

2. **Inner loop** (up to 80 sweeps): coordinate descent over all 15 parameters. Each parameter is optimised by a multi-pass grid search (4 passes $\times$ 100 grid points, geometrically refining the search window).

Best-checkpoint selection. The weighted loss $\mathcal{L}$ monotonically increases across outer iterations (because weights grow), but the *unweighted* RMS error may degrade after the optimal reweighting round. The fitter tracks the unweighted RMS at every outer iteration and returns the parameters from the checkpoint with the lowest unweighted RMS, not the final iterate.

Chapter 6

Fitted Parameters (v2.6.2)

Parameter	Value	Physical role
$k_p^{(1)}$	20.820	Period 1 (H, He)
$k_p^{(2)}$	5.381	Period 2 (Li–Ne)
$k_p^{(3)}$	5.117	Period 3 (Na–Ar)
$k_p^{(4)}$	3.956	Period 4 (K–Kr)
$k_p^{(5)}$	3.728	Period 5 (Rb–Xe)
$k_p^{(6)}$	2.919	Period 6 (Cs–Rn)
$k_p^{(7)}$	2.665	Period 7 (Fr–Og)
a_{soft}	-0.680	Softness coupling
b_χ	0.225	Electronegativity exponent
$c_b^{(s)}$	1.464	s-block multiplier
$c_b^{(p)}$	0.973	p-block multiplier
$c_b^{(d)}$	1.605	d-block multiplier
$c_b^{(f)}$	1.745	f-block multiplier
c_{shell}	0.328	Shell-closure suppression amplitude
a_{rel}	0.054	Relativistic stiffening amplitude

The monotonic decrease $k_p^{(1)} > k_p^{(2)} > \cdots > k_p^{(7)}$ reflects increasing nuclear shielding: each additional shell of core electrons makes the volume-to-polarizability conversion less efficient. The d-block multiplier $c_b^{(d)} \approx 1.6$ confirms that partially-filled d shells contribute significant polarisable density beyond what r_{cov}^3 alone predicts. The shell-closure amplitude $c_{\text{shell}} \approx 0.33$ suppresses noble-gas predictions by a third, while the relativistic amplitude $a_{\text{rel}} \approx 0.05$ provides a targeted $\sim 5\%$ correction at the Hg hotspot.

Chapter 7

Validation Results

7.1 Global Accuracy

Metric	**Z = 1–118**	**Z = 1–54**
RMS error	17.5%	14.8%
Max error	64.5% (Lv)	57.4% (Ne)
Monotonicity breaks	10/70 pairs	5/35 pairs
Elements >20% error	23/118	8/54

7.2 Per-Block RMS

Block	**RMS**	**Physical note**
s-block	19.0%	Alkali/alkaline-earth: large α, well constrained
p-block	19.9%	Noble-gas suppression now active (g_{shell})
d-block	15.5%	Transition metals: $c_b^{(d)}$ absorbs d-shell effect
f-block	16.0%	Lanthanides/actinides: fixed $n_{\text{val}} = 3$ works well

7.3 Improvement from v2.6.0 to v2.6.2

Regime	**v2.6.0**	**v2.6.2**	**Improvement**
Overall RMS	19.1%	17.5%	−8.4%
Noble gases RMS	36.7%	27.3%	−25.6%
Halogens RMS	24.1%	14.2%	−41.1%
Period 6 RMS	15.4%	14.1%	−8.4%
Hg error	+36.6%	+34.0%	−7.1%
Ne error	+85.0%	+57.4%	−32.5%
Max error	85.0%	64.5%	−24.1%

7.4 Known Failure Modes

1. **Neon** ($\alpha_{\text{ref}} = 0.396$ Å^3, error +57%): Ne has an anomalously small polarizability relative to its covalent radius. The model's χ-based suppression is insufficient because $r_{\text{cov}}(\text{Ne}) = 0.67$ Å overestimates the true electron-cloud extent for this closed-shell atom.
2. **Nobelium** ($\alpha_{\text{ref}} = 17.5$ Å^3, error +48%): No has strong relativistic contraction of the $7s$ orbital that reduces α below the r_{cov}^3 expectation. Our non-relativistic descriptors cannot capture this.
3. **Superheavy p-block** (Lv, Ts: errors >40%): Extreme spin–orbit coupling in the $7p_{1/2}$ / $7p_{3/2}$ manifold changes the effective orbital extent. Relativistic CCSD(T) reference values carry significant uncertainty themselves.

Chapter 8

Integration with the SCF Polarization Solver

The alpha model provides the atomic polarizabilities $\alpha_i = \alpha(Z_i)$ that enter the Applequist–Thole SCF equations:

$$\boldsymbol{\mu}_i = \alpha_i \, \mathbf{E}_i^{\mathrm{loc}} \tag{8.1}$$

$$\mathbf{E}_i^{\mathrm{loc}} = \mathbf{E}_i^{\mathrm{perm}} + \sum_{\substack{j \neq i \\ r_{ij} > r_{\mathrm{excl}}}} f(u_{ij}) \, \mathbf{T}_{ij} \, \boldsymbol{\mu}_j \tag{8.2}$$

where $\mathbf{T}_{ij}$ is the dipole–dipole interaction tensor and $f(u)$ is the Thole exponential damping function:

$$f(u) = 1 - \left(1 + \tfrac{a\,u}{2}\right) e^{-au}, \qquad u_{ij} = \frac{r_{ij}}{(\alpha_i \, \alpha_j)^{1/6}}, \qquad a = 2.6 \tag{8.3}$$

Unit convention. All electric fields ($\mathbf{E}^{\mathrm{perm}}$ from charges, $\mathbf{T}_{ij}\boldsymbol{\mu}_j$ from dipoles) are computed in Gaussian units ($e/\text{Å}^2$) without the Coulomb constant k_e. The constant $k_e = 332.0636$ kcal $\cdot$ Å/(mol $\cdot$ e^2) appears only in the energy formula:

$$U_{\mathrm{pol}} = -\frac{1}{2} \, k_e \sum_{i=1}^{N} \boldsymbol{\mu}_i \cdot \mathbf{E}_i^{\mathrm{perm}} \tag{8.4}$$

Exclusion radius. Pairs with $r_{ij} < r_{\mathrm{excl}} = 1.6$ Å are excluded from both the permanent-charge driving field and the dipole–dipole tensor. This is equivalent to standard 1-2/1-3 exclusions in AMOEBA without requiring bond topology.

Chapter 9

Nuclear Stability Integration

The polarizability model is tied into a nuclear stability module (`nuclear_stability.hpp`) that silently annotates each prediction with a confidence score derived entirely from atomic structure.

9.1 Ab-Initio Stability from Z

Every call to `alpha_predict_checked(Z)` runs a silent stability check using three physics engines—no external decay data:

1. **Semi-empirical mass formula** (Bethe–Weizsäcker, 5 parameters + shell correction): Predicts binding energy $B(Z, A)$, most stable mass number A^*, and the fissility parameter Z^2/A.
2. **Nuclear shell model** (magic numbers 2, 8, 20, 28, 50, 82, 126): Gaussian shell correction to B enhances stability at closed proton/neutron shells.
3. **Geiger–Nuttall law** (Viola–Seaborg parameterisation): Estimates α-decay half-life from the Q-value $Q_\alpha = B(Z{-}2, A{-}4) + B(2, 4) - B(Z, A)$.

9.2 Stability Classes and Confidence

Class	Criterion	Conf.	Example
Stable	$\exists$ stable isotope	1.00	C, Fe, Pb
PrimordialLong	$t_{1/2} > 4.5\,\mathrm{Gy}$	0.85	Th, U, Bi
Radioactive	$t_{1/2} > 1\,\mathrm{s}$	0.47–0.90	Tc, Pm, Cf
VeryShortLived	$t_{1/2} \in [1\,\mu\mathrm{s}, 1\,\mathrm{s})$	0.10–0.30	—
Superheavy	$t_{1/2} < 1\,\mathrm{ms}$ or $Z \geq 104$	0.02–0.25	Cn, Lv, Og

The confidence score gates the AlphaCalibrator's acceptance threshold: stable elements use the strict 1% gate; superheavy elements widen to 20% (acknowledging that their polarizabilities are purely theoretical).

9.3 Known Limits

- **He-4**: Doubly magic (Z=2, N=2). Hardcoded $A^* = 4$ because the SEMF liquid-drop model cannot resolve this light nucleus.
- **Tc, Pm**: Correctly identified as having no stable isotopes despite $Z < 83$; classified as Radioactive (β^- decay).
- **Z=104–118**: Classified Superheavy with confidence ≤ 0.25. Half-lives are clamped at $\leq 10^5$ s to prevent SEMF over-prediction of stability for transactinides.

Bibliography

[1] K. J. Miller, "Additivity methods in molecular polarizability," *J. Am. Chem. Soc.* **112**, 8533 (1990).

[2] P. Schwerdtfeger and J. K. Nagle, "2018 Table of static dipole polarizabilities of the neutral elements in the periodic table," *Mol. Phys.* **117**, 1200 (2019).

[3] P. Pyykkö and M. Atsumi, "Molecular single-bond covalent radii for elements 1–118," *Chem. Eur. J.* **15**, 186 (2009).

[4] J. Applequist, J. R. Carl, and K.-K. Fung, "An atom dipole interaction model for molecular polarizability," *J. Am. Chem. Soc.* **94**, 2952 (1972).

[5] B. T. Thole, "Molecular polarizabilities calculated with a modified dipole interaction," *Chem. Phys.* **59**, 341 (1981).

Part II

Plateau Investigation

Part C — v2.6.3 and v2.7.0 Experiments, Root-Cause Analysis, and Reversion to v2.6.2

Chapter 10

Background and Baseline

Parts A and B (Part I of this booklet) established the v2.6.2 model with **15 parameters**:

```
alpha(Z) = c_block(Z) * k_period(Z) * r(Z)^3
           * (1 + a_soft * s(Z)) / chi(Z)^b_chi
           * g_shell(Z)
           * g_rel(Z)

g_shell(Z) = 1 - c_shell * shell_closure(Z)   [shell-closure gate]
g_rel(Z)   = 1 - a_rel   * f_rel(Z)           [relativistic stiffening]
```

The v2.6.2 baseline performance was:

Metric	**Value**
RMS error (Z=1–118)	17.5%
RMS error (Z=1–54)	14.8%
Noble gas RMS	27.3%
Max error	64.5% (Lv)
Monotonicity breaks	10/70

The goal entering the v2.6.3 / v2.7.0 cycle was to push the global RMS below **15%**, with particular focus on eliminating the known failure regimes: noble gases, the Li/Ne volume degeneracy, and f-block scatter.

Chapter 11

Experiment 1 — v2.6.3: Period-Modulated Shell Closure

11.1 Motivation

The v2.6.2 shell-closure gate applies a **uniform suppression** to noble gases regardless of period. The physical expectation is that heavier noble gases (Kr, Xe, Rn) should be *less* suppressed than light ones (He, Ne) because their outer electrons are farther from the nucleus and more deformable. The suppression appropriate for He (period 1) over-penalises Rn (period 6).

11.2 Implementation

Added one parameter `c_shell_decay` (16 parameters total). For noble gases, the shell-closure proximity ϕ_{eff} was period-modulated:

```
phi_eff(Z) = exp(-c_shell_decay * (period(Z) - 1))   [noble gases only]
phi_eff(Z) = shell_closure(Z)                         [all other
   elements]
```

This decouples the suppression amplitude for He/Ne from that for Kr/Xe/Rn.

11.3 Result

Metric	**v2.6.2**	**v2.6.3**	Δ
RMS (Z=1–118)	17.5%	≈17.3%	−0.2%
Noble gas RMS	27.3%	≈24.1%	−3.2%
Overall	—	—	marginal

The noble-gas sub-regime improved modestly. However, the global RMS barely moved. The `c_shell_decay` parameter coupled strongly with `k_period[1]` (period 1, H and He): lowering the suppression on heavier noble gases forced `k_period[1]` and `c_shell` to re-compensate for He and Ne, erasing most of the gain elsewhere. The parameter was not independently informative enough to justify the added complexity. **Experiment discarded.**

Chapter 12

Experiment 2 — v2.7.0: Fill-Fraction Volume Correction

12.1 Motivation

A known structural deficiency of the v2.6.x base model is that $r_{\text{cov}}(Z)^3$ cannot simultaneously account for:

- **Alkali metals (Li, Na, K . . .)**: one electron in a partially-filled shell, diffuse cloud, *underestimated* by the model (-25 to -33%).
- **Noble gases (He, Ne)**: full shells, compact cloud, *overestimated* even after g_{shell} suppression ($+30$ to $+57\%$).

Both errors arise from the same cause: r_{cov} measures a bonding half-distance, not the actual valence-shell radius. The fill fraction $f = n_{\text{val}}/n_{\text{max}}$ is a proxy for how "expanded" or "contracted" the outer shell is relative to r_{cov}. This idea was formalised as:

```
g_fill  = 1 + c_orb * (0.5 - fill_fraction(Z))
r_eff^3 = rcov^3 * max(g_fill, 0.1)
```

- Sparse shells (Li: $f = 0.125$): $g_{\text{fill}} > 1 \Rightarrow$ larger effective volume
- Full shells (Ne: $f = 1.0$): $g_{\text{fill}} < 1 \Rightarrow$ smaller effective volume
- Mid-shell anchor (C, Si: $f = 0.5$): $g_{\text{fill}} = 1 \Rightarrow$ unchanged

12.2 Implementation

Added `c_orb` (17 parameters total). The `fill_fraction(Z)` descriptor was already present in `atomic_descriptors.hpp`. The model was re-fitted with the full 17-parameter coordinate descent.

12.3 Result

Metric	**v2.6.3**	**v2.7.0**	**Δ**
RMS (Z=1–118)	≈17.3%	$17.02 \pm 0.04\%$	−0.3%

The model plateaued at ≈17.02% and did not improve further across outer iterations. This is the plateau event that motivated Part C. **Experiment discarded.**

Chapter 13

Root-Cause Analysis of the 17% Plateau

Four independent causes stack to produce the observed floor.

13.1 Cause 1 — f-block Descriptor Degeneracy (primary, ∼6–8% irreducible)

The design decision in `atomic_descriptors.hpp` (Section 4.3) fixes `active_valence = 3` for **all** lanthanides (Z=57–71) and actinides (Z=89–103). This was intentional: it prevents an artificial monotonic gradient in softness across each series.

The consequence is that `fill_fraction(Z)`, `softness(Z)`, and `chi(Z)` are **nearly constant across the entire lanthanide series**. The only intra-series variation is from $r_{\mathrm{cov}}(Z)$, which spans only ≈11% (1.62–1.80 Å) across La–Lu, giving a r^3_{cov} range of ≈37%.

The reference data spans **≈60%** (La=31.1 Å^3 → Tm/Yb=19.5 Å^3) and is **non-monotonic**:

Element	**Z**	α_{ref} **(Å^3)**	**Anomaly**
La	57	31.1	—
Ce	58	29.6	—
Pr	59	28.2	—
Nd	60	31.4	← bump above La ($4f$ anomaly)
Sm	62	23.8	—
Eu	63	27.7	← half-filled $4f$ shell bump
Yb	70	19.5	← filled $4f$ shell contraction

These anomalies are real $4f$ physics that no smooth $r^3_{\mathrm{cov}} \cdot c^{(f)}_{\mathrm{block}}$ trend can represent.

Conclusion. The ≈17% floor is largely set by the f-block and cannot be broken without either relaxing the `active_valence = 3` constraint or adding an explicit $4f$-anomaly correction term.

13.2 Cause 2 — Objective/Metric Mismatch

The fitter minimises a **log-space Huber loss**:

```
L(theta) = sum w_i * Huber(log(alpha_pred) - log(alpha_ref))
```

But the reported performance metric is a **linear-space RMS % error**:

```
pct_i = 100 * (alpha_pred - alpha_ref) / alpha_ref
RMS   = sqrt( sum(pct_i^2) / N )
```

These are different norms. A 5% log-space error on Cs ($\alpha = 59.4$ Å^3) is "small" to the optimiser but contributes a large absolute residual to the linear RMS%. The practical consequence: **the optimiser can legitimately converge in log-space while the linear RMS% plateau is driven by moderate-log-error, high-α elements** (K, Rb, Cs, Ba). Adding more parameters cannot resolve this without also changing what the optimiser minimises.

13.3 Cause 3 — Coordinate Descent Saddle in a Collinear Parameter Space

The parameters `k_period[p]`, `c_block[b]`, and `c_orb` all multiply the same r_{eff}^3 term:

```
alpha  proportional to  c_block[b] * k_period[p] * r_eff^3
```

These are **collinear in their effect on the objective**: increasing `k_period[2]` by 10% has the same effect as increasing `c_block[s]` by 10% for any period-3 s-block element. Coordinate descent cannot make progress along a collinear direction. Gradient-based methods (L-BFGS, Adam) would use curvature information to escape; the current grid-scan CD has no such mechanism.

13.4 Cause 4 — Smoothness Penalty vs. Real Non-Monotonic Physics

The group-pair smoothness penalty (`eta_smooth = 0.15`) penalises cases where the model's group trend disagrees with the reference:

```
L_smooth = eta * sum_{(a,b) in group pairs} Huber(Delta_pred - Delta_ref
   )
```

This was designed to prevent erratic per-element fits that break periodic trends. However, the **lanthanide series contains genuine non-monotonic bumps** (Nd, Eu) that look like noise to the penalty but are real $4f$ physics. The penalty actively pushes the optimiser *away* from fitting these elements correctly. Since $c_b^{(f)}$ is a single multiplier for all 15 f-block elements, the only solution available to the model is a smooth r_{cov}^3 curve that necessarily misses Nd and Eu.

Chapter 14

Quantitative Decomposition of the 17% Floor

Source	**Est. RMS contribution**	**Breakable with current arch.?**
f-block descriptor degeneracy	6–8%	Only with architectural change
High-α alkali metric mismatch	2–4%	Partly (change reported metric)
Correlated parameter saddle (CD)	2–4%	Yes (gradient-based optimiser)
Smoothness penalty vs. f-anomalies	2–3%	Partly (exclude f-block from penalty)
`c_orb` insufficient dynamic range	1–2%	Yes (wider range, separate s/p terms)
Reducible floor (current descriptors)	**≈12–13%**	—
Irreducible floor (f-block physics)	**≈10–12%**	—

Chapter 15

Decision: Revert to v2.6.2

Given the analysis above:

- **v2.6.3** (`c_shell_decay`): marginal improvement (-0.2% global RMS), adds parameter coupling complexity. **Discarded.**
- **v2.7.0** (`c_orb`): marginal improvement (-0.5% global RMS across both experiments), does not address the dominant failure mechanism. **Discarded.**

The code has been reverted to **v2.6.2** (15 parameters, RMS $= 17.5\%$):

- `alpha_model.hpp`: `AlphaModelParams` restored to v2.6.2 values; `alpha_predict` simplified back to direct `shell_closure(Z)` gate.
- `fit_alpha_model.cpp`: `cd_sweep` and JSON writer stripped of `c_orb` and `c_shell_decay`.

v2.6.2 is the current stable baseline. It is well-characterised, reproducible, and its failure modes are fully documented. No further incremental parameter additions should be attempted without first addressing at least one of the four root causes in Chapter 13.

Chapter 16

Anti-Pattern Log

The following approaches have been tried and failed to break the plateau. They should **not** be retried without addressing the underlying cause first.

Approach	Version	Why it failed
Period-decay of noble-gas suppression	v2.6.3	Coupled to `k_period[1]`; He/Ne compensation erased g
Fill-fraction volume correction	v2.7.0	f-block fill fraction is constant; collinear with `k_period`
More outer reweighting iterations (30→50)	v2.7.0	Weight saturation plateau; best checkpoint still ≈17%
Wider coordinate descent grid	ongoing	Does not escape correlated-parameter saddles

Chapter 17

Conclusion

The v2.6.3 and v2.7.0 experiments were a failure in the narrow sense that neither broke the 17% RMS floor. They were a success in the broader sense that they conclusively characterised *why* the floor exists and established that it cannot be broken by adding parameters to the current descriptor architecture. The four-cause analysis in Chapter 13 and the anti-pattern log in Chapter 16 are the durable outputs of this work: they prevent the same dead ends from being explored a second time.

The generative model was always a pragmatic compromise—a way to obtain a reasonable $\alpha(Z)$ for any element using only atomic number, with no external data at runtime. That constraint made it impossible to represent the genuine non-monotonic $4f$ physics in the lanthanide series, the anomalous compactness of Ne relative to its covalent radius, or the explicit relativistic contractions of the superheavy block.

Going forward: empirical values. For production use, $\alpha(Z)$ will be set from the empirical reference table directly—the same 118-entry dataset (Miller 1990, Schwerdtfeger & Nagle 2019) that was used to fit the generative model. This eliminates all model error for the 118 tabulated elements and removes any motivation for further parameter tuning. The generative model (`alpha_model.hpp`) is retained as a fallback for elements outside the table (Z = 119–120) and as a documented record of what a closed-form predictor can and cannot achieve.

Chapter 18

Part D — Empirical Lookup Table

Direction: Replace the generative model prediction with a direct lookup into the 118-entry empirical reference table for $Z = 1$–118. The generative model is kept as the fallback path for $Z > 118$.

Mechanism: At runtime, `alpha_predict(Z)` checks whether $Z \leq 118$. If so, it returns the tabulated value from a compile-time constant array (same source data as `polarizability_ref.csv`). If $Z > 118$, it falls through to the existing generative equation with $k_p^{(7)}$ as the period-7 floor.

Expected outcome: RMS error for $Z = 1$–118 drops to 0% by construction. The nuclear-stability confidence scores remain active and continue to annotate each returned value. The offline fitter (`fit_alpha_model.cpp`) becomes a diagnostic tool rather than a production dependency.

T1 invariant impact: None. The `active_valence = 3` invariant in `atomic_descriptors.hpp` is a descriptor-layer constraint and is unaffected by switching the prediction path.

Chapter 19

Document Index

Document	Content
`docs/section_alpha_model.tex`	Parts A/B source (model derivation, v2.6.0–v2.6.2)
`docs/ALPHA_MODEL_REPORT_PARTC.md`	Part C source (plateau investigation)
`docs/ALPHA_MODEL_BOOKLET.tex`	**This file** (compiled booklet, Parts A–C)
`docs/ALPHA_MODEL_REPORT_PARTD.md`	Part D: empirical lookup table (this chapter)
`atomistic/models/alpha_model.hpp`	Active model code (v2.6.2)
`atomistic/models/atomic_descriptors.hpp`	Descriptor layer (unchanged)
`tools/fit_alpha_model.cpp`	Offline fitter (reverted to v2.6.2)
`config/alpha_model_params.json`	Saved parameter checkpoint
`tests/test_alpha_model.cpp`	T1–T4 invariant tests
`tests/test_alpha_calibrator.cpp`	AlphaCalibrator tests